

Lab 5

NOTE THAT THE **IOR effect (cued - uncued)** IN THE LAB DATA FOR ONDREJ IS WRONG! IT SHOULD BE -7

Q1.

1)

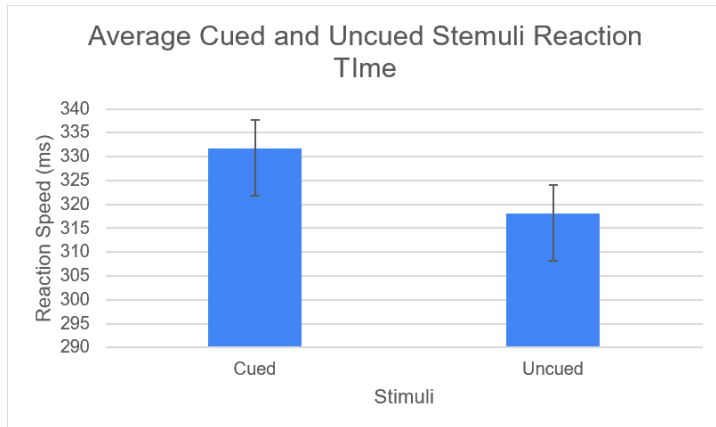


Figure 1: Comparing Average Cued and Uncued Stimuli Reaction Time bar graph using Lab 5 content with custom SEM bars.

2)

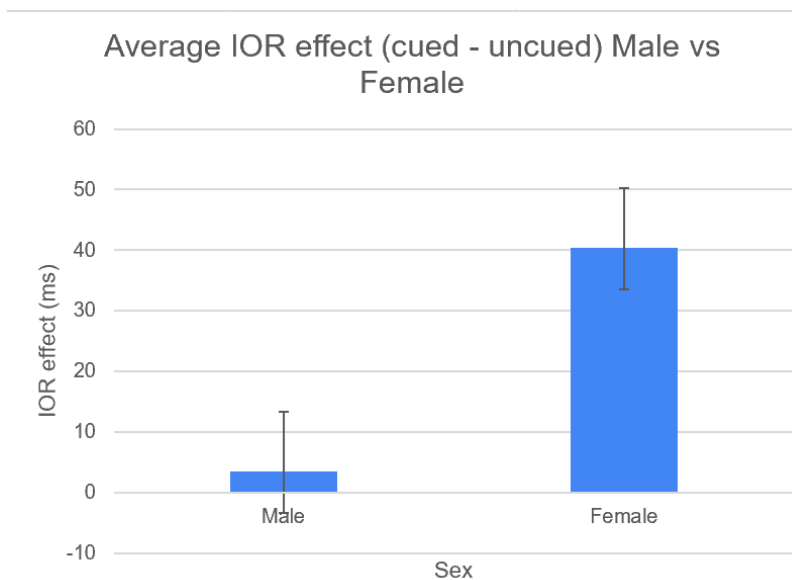


Figure 2: Comparing Average IOR effect (cued – uncued) Male vs Female bar graph using Lab 5 content with custom SEM bars.

Q2.

Q2.

Step 1 - Null hypothesis (H_0): $\mu_{\text{Cued}} = \mu_{\text{Uncued}}$
 Alternative hypothesis (H_1): $\mu_{\text{Cued}} \neq \mu_{\text{Uncued}}$

Step 2 - $\alpha = 0.05$

Step 3 - $df = n - 1$
 $df = 14 - 1 = 13$

Step 4 - Critical value for $df = 13$ @ $\alpha = 0.05$
 (two tailed) is 2.16
 Reject H_0 if absolute t-stat > 2.16

Step 5 - $t = \frac{\bar{X}_D}{S_D / \sqrt{n}}$

$\bar{X}_D = 13.64$

$S_D = \sqrt{\frac{\sum X^2 - \frac{(\sum X)^2}{n}}{n-1}} = \sqrt{\frac{1912 - \frac{14^2}{13}}{13}} = 30.94$

$t = \frac{13.64}{30.94 / \sqrt{14}} = 1.649$
 $t \approx 1.65$

∴ Final t-stat is 1.65

t-Test: Paired Two Sample for Means

	Cued Conditions (ms)	Uncued Conditions (ms)
Mean	331.7142857	318.0714286
Variance	491.7582418	1387.148352
Observations	14	14
Pearson Correlation	0.55782041	
Hypothesized Mean D	0	
df	13	
t Stat	1.649699879	
P(T<=t) one-tail	0.061472405	
t Critical one-tail	1.770933396	
P(T<=t) two-tail	0.12294481	
t Critical two-tail	2.160368656	

Figure 3: t-test paired two sample for means for Cued and Uncued Conditions.

Q3.

t-Test: Two-Sample Assuming Equal Variances		
	M	F
Mean	-1.222222222	40.4
Variance	745.1944444	229.3
Observations	9	5
Pooled Variance	573.2296296	
Hypothesized M	0	
df	12	
t Stat	-3.11675875	
P(T<=t) one-tail	0.004454145	
t Critical one-tail	1.782287556	
P(T<=t) two-tail	0.00890829	
t Critical two-tail	2.17881283	

Figure 4: t-test Two Sample Assuming Equal Variances for Male and Female.

Q3. Null Hypothesis (H_0): $\mu_{\text{Male}} = \mu_{\text{Female}}$
Alternative Hypothesis (H_1): $\mu_{\text{Male}} \neq \mu_{\text{Female}}$

$\alpha = 0.05$

$df = (n_1 - 1) + (n_2 - 1)$
 $9 - 1 + 5 - 1 = 12$

For $df = 12$ @ $\alpha = 0.05$ (two-tailed), the critical value is 2.179

Reject H_0 if $t < -2.179$ or > 2.179

$S_p^2 = \frac{SS_1 + SS_2}{df_1 + df_2} = \frac{7043.32 + 917.20}{9 + 5}$

$S_p^2 = 663.39$

$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{-1.22 - 40.40}{\sqrt{663.39 \left(\frac{1}{9} + \frac{1}{5} \right)}}$

$t = -2.90$ $\therefore$ Our t of -2.90 is outside critical range (± 2.179)

Q4.

- a) A paired samples t-test as performed and found to be not significant ($t=1.65$, $df=13$, $p=0.12$). The Cued group mean (331.71 ms) was the same as the Uncued group mean

(318.07 ms). While the Cued mean was numerically higher (slower) because the p-value of 0.12 is greater than the alpha of 0.05, the statistical conclusion is that there is no significant difference.

- b) An Independent samples t-test was performed and found to be significant ($t=-2.90$, $df=12$, $p=0.01$). The Female group (40.40 ms) had a larger IOR effect than the Male group (-1.22 ms). The Females showed a much larger difference between cued and uncued conditions than the males.
- c) Independent Variable: The cue condition (Cued vs. Uncued stimuli). This is the variable you are changing between conditions
Dependent Variable: The response time measured in milliseconds (ms). This is what you are measuring.
- d) The Hypothesis relates to the IOR mechanism that was described in the instructions.
Null Hypothesis (H_0): There is no significant difference in response times between cued and uncued conditions; any observed difference is due to chance.
Alternative Hypothesis (H_1): There is a significant difference in response time between conditions, specifically that response times will be slower for cued conditions due to the IOR effect.

Q5.

t-Test: Paired Two Sample for Means		
	<i>Trial 1</i>	<i>Trial 10</i>
Mean	361.75	613
Variance	16728.56818	35184.36364
Observations	12	12
Pearson Correl:	0.145551218	
Hypothesized μ	0	
df	11	
t Stat	-4.109724589	
P(T<=t) one-tail	0.000865297	
t Critical one-ta	1.795884819	
P(T<=t) two-tail	0.001730594	
t Critical two-tai	2.20098516	

Figure 5: t-Test Paired Two Sample for Means from Lab 2 using Trial 1 and 10.

Q5

$$\sum X = -3015$$

$$\sum X^2 = 1,250,875$$

$$SD = \sqrt{\frac{1,250,875 - \frac{(-3015)^2}{12}}{12-1}}$$

$$= 211.78$$

A paired-samples t-test was performed and found to be significant ($t = -4.11$, $df = 11$, $p = 0.0017$). The Trial 10 group mean of 613 was significantly higher (better) than the Trial 1 group mean of 361.75.

Q6.

Size of each group	t-value	D.F (df)	Significant
16, 15	2.14	29	$p < 0.05$
24, 24	2.18	46	$p < 0.05$
12, 11	2.04	21	N/S ($p > 0.05$)
9, 16	3.01	23	$p < 0.05$

Verifying:

Row 1 ($df = 29$): critical value for df @ $\alpha = 0.05$ is approx 2.045. Since t-value is 2.14 (higher), it is significant.

Row 2 ($df = 46$): critical value for df @ $\alpha = 0.05$ is approx 2.01. Since t-value is 2.86 (higher), it is significant.

Row 3 ($df = 21$): critical value for df @ $\alpha = 0.05$ is approx 2.08. Since t-value is 2.04 (lower), it is NOT significant.

Row 4 ($df = 23$): critical value for df @ $\alpha = 0.05$ is approx 2.069. Since t-value is 3.01 (higher), it is significant.

QUESTION 7 BELOW:

Q7.

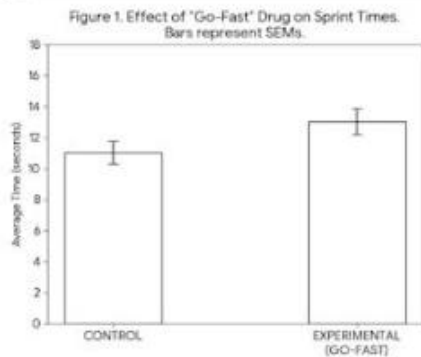


Figure 1: Effect of Go-Fast Drug on Sprint Times. Bars represent SEMs.

a)

Variables:

Independent variable is the drug "Go-Fast", more specifically, whether the sprinter received the drug or was in the control group.

Dependent Variable: The time it takes to run the 100 meters.

b)

For the precaution, the coach should have used random assignment when dividing the subjects into groups. This makes sure that any individual differences like natural speed and/or fitness levels, are distributed evenly across both groups in order to prevent bias.

c)

$$T = 2 + 0.736632 = 2.71506$$

Based on the calculations, an independent samples t-test yielded a t-value of 1.81 with 33 degrees of freedom. The resulting p-value was .079, which is greater than the standard significance level of .05.

Results Statement (Modeled on Example)

Means for the dependent measure (100m sprint time) were calculated for each group and these results are illustrated in Figure 1. An independent t-test was carried out to compare the groups, and the "Go-Fast" group failed to show a significant increase in speed compared to the control group ($t=2.71$; $df=33$; $p>.05$). In fact, the mean time for the experimental group ($M=13.0$) was numerically slower than the control group ($M=11.0$). It is evident from Figure 1 that the drug did not have the intended effect, and the observed difference was not statistically significant.

Conclusion: The coach can conclude that "Go-Fast" does not increase the speed of his sprinters. If anything, the sprinters who took the drug were slower on average, although the statistical test suggests this difference could be due to chance.

